

Enhancing Predictive Accuracy for Agricultural Crop Yields in Indian States

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Abstract —Agricultural productivity is critical for the sustenance and economic growth of India, a country with diverse climatic regions and crop practices. Accurate prediction of crop yields is essential for optimizing agricultural outputs, ensuring food security, and planning resource allocation. Traditional predictive models often struggle with the non-linear and complex nature of factors influencing crop yields. This study introduces an innovative approach by integrating power transformations, specifically the Yeo-Johnson transformation, within machine learning models to enhance the predictive accuracy of agricultural crop yields in various Indian states. We collected a comprehensive dataset comprising climatic variables, soil properties, crop types, and historical yield data across different Indian states. The Yeo-Johnson power transformation was applied to normalize the distribution of these features, addressing issues of skewness and heteroscedasticity, thereby making the data more suitable for machine learning algorithms. We evaluated several machine learning models, including Linear Regression, Random Forest, and Gradient Boosting, to establish a benchmark for comparison. Our findings demonstrate a significant improvement in predictive accuracy with the application of the Yeo-Johnson transformation across all models tested are shown in table 1. The Gradient Boosting model, post-transformation, exhibited the highest accuracy, underscoring the potential of combining power transformations with ensemble learning techniques for crop yield prediction. This approach not only aids in achieving greater precision in forecasts but also provides insights into the relative importance of various factors affecting crop yields. The study emphasizes the role of advanced data preprocessing techniques, such as power transformations, in unlocking the full potential of machine learning for agricultural applications. By enhancing the predictive accuracy of crop yield models, stakeholders can make more informed decisions, leading to improved agricultural productivity and sustainability in the diverse and challenging landscape of Indian agriculture.

Keywords— Agricultural Crop Yields, Predictive Accuracy, Power Transformation, Crop Yield Prediction, Ensemble Learning Techniques, Climatic Variables in Agriculture, Agricultural Data Analysis

I. Introduction

Agriculture stands as a cornerstone of India's economy, not only as a significant contributor to its GDP but also as a primary source of livelihood for over half of the country's population. The diverse climatic zones across Indian states result in a wide variety of crops; however, this diversity also introduces complexities in predicting agricultural outputs. Accurate crop yield predictions are vital for ensuring food security, optimizing resource allocation, and enhancing agricultural planning and policymaking. Traditional approaches to predicting agricultural yields have relied heavily on historical yield data, climatic conditions, and soil characteristics. Yet, these models often fall short in capturing the non-linear relationships and heterogeneity inherent in agricultural data. The advent of machine learning (ML) offers promising avenues to overcome these limitations by leveraging complex algorithms capable of identifying intricate patterns in large datasets. [1-5] Despite the advancements, ML models' performance in agricultural predictions can be significantly affected by the distribution of the input data. Skewed or non-normally distributed data can bias predictions, leading to inaccurate yield forecasts. Power transformations, particularly the Yeo-Johnson transformation, have emerged as powerful techniques to address these issues, transforming data towards normality and stabilizing variance, thereby enhancing model performance. This study aims to investigate the effectiveness of incorporating the Yeo-Johnson power transformation within ML models to improve the predictive accuracy of crop yields in Indian states. [6-10] By comparing the performance of several ML models, both with and without the application of power transformations, this research seeks to identify optimal strategies for yield prediction in the complex and varied context of Indian agriculture. Moreover, the study contributes to the broader field of agricultural informatics by demonstrating the potential of advanced data preprocessing techniques in enhancing the utility of ML for agricultural applications. The findings are expected to offer valuable insights for agronomists, policymakers, and data scientists working towards the goal of sustainable and efficient agricultural practices in India.

II. Principle and Working

The principle behind enhancing predictive accuracy for agricultural crop yields using machine learning (ML) techniques, particularly with the integration of power transformations like the Yeo-Johnson transformation,

revolves around preparing and processing the data to better suit the assumptions of ML algorithms. This section explores the theoretical underpinnings of these methods and elucidates their practical application in the context of agricultural data analysis.

A. Machine Learning in Agriculture: At its core, machine learning involves training algorithms to recognize patterns and make predictions or decisions based on data. In agriculture, ML can identify complex relationships between various factors (e.g., climate, soil properties, crop types) and crop yields. The primary principle is to leverage historical data to forecast future agricultural outputs accurately.

B. Power Transformations: The Yeo-Johnson transformation is a power transformation method that aims to make data more normally distributed. Many ML algorithms assume that input features are normally distributed, making such transformations crucial for improving model performance. The Yeo-Johnson transformation, unlike the Box-Cox transformation, can handle both positive and zero (or negative) values, making it more versatile for different types of data.

C. Yeo-Johnson Transformation: Once the data is preprocessed, the Yeo-Johnson transformation is applied to features not following a normal distribution. This process adjusts each feature to approximate a normal distribution, thereby stabilizing variance and minimizing skewness. The transformation is defined as follows for any feature X :

$$X_{transformed} = \begin{cases} [(X+1)^\lambda - 1]/\lambda & \text{if } \lambda \neq 0, X \geq 0 \\ \ln(X+1) & \text{if } \lambda = 0, X \geq 0 \\ -[(-X+1)^{2-\lambda} - 1]/(2-\lambda) & \text{if } \lambda \neq 2, X < 0 \\ -\ln(-X+1) & \text{if } \lambda = 2, X < 0 \end{cases}$$

The integration of Yeo-Johnson power transformations within ML models for predicting agricultural crop yields is predicated on the principle of data normalization to improve algorithm performance. The working process encompasses data preparation, transformation, model training and evaluation, and ultimately, the deployment of optimized models for accurate yield predictions.

III . Block Diagram

A. Block for Power Transformations:

The Power Transformations block, specifically focusing on the Yeo-Johnson method, plays a critical role in the preprocessing phase of a machine learning pipeline aimed at predicting agricultural crop yields. At this stage, the preprocessed dataset, which has already been cleaned and initially prepared (including steps like handling missing values, encoding categorical variables, and basic feature selection), is evaluated for the distribution of its features. Many machine learning algorithms assume that input features follow a normal or Gaussian distribution, which is often not the case in real-world data, especially in

agriculture, where data can be highly skewed due to various factors such as climate variations, soil types, and farming practices. [11-15] The process begins with the identification of skewed features within the dataset. Skewness is a measure of the asymmetry of the probability distribution of a real-valued random variable are shown Fig.1. Significantly skewed features can lead to biases in model training and ultimately affect the predictive accuracy of the machine learning models. Once these features are identified, the Yeo-Johnson transformation is applied. The Yeo-Johnson method is particularly suited for this task as it supports both positive and negative values, unlike some other transformations that only deal with positive values. This makes it a versatile choice for agricultural datasets that might contain a wide range of values across different features. The transformation itself adjusts the distribution of each identified feature towards a more Gaussian-like shape, thereby stabilizing variance and minimizing skewness. This step is crucial for enhancing model performance, as it aligns the data closer to the underlying assumptions of many machine learning algorithms. By transforming the data in this manner, the subsequent training of machine learning models can proceed on a dataset that is more likely to produce accurate and reliable predictive results.

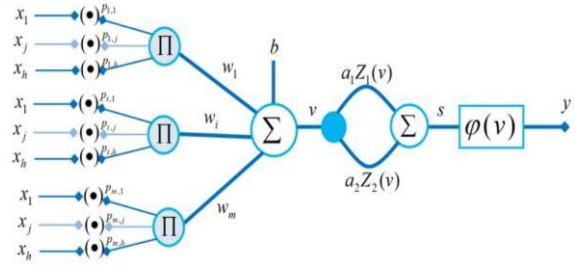


Fig.1 Block Diagram for power transformation

IV. Proposed Method

The proposed model for enhancing the predictive accuracy of agricultural crop yields integrates the use of power transformations, specifically the Yeo-Johnson method, within a comprehensive machine learning framework to address the complexities inherent in agricultural data. This model is meticulously designed to navigate the challenges posed by the variability and skewness of data, thereby aiming to yield more accurate and reliable predictions for crop yields. At the outset, the model emphasizes rigorous data collection and aggregation, gathering diverse datasets that include weather conditions, soil properties, historical yield data, and potentially remote sensing data, to form a robust foundation for analysis. The initial phase of data preprocessing involves cleaning, encoding, and normalizing data to prepare it for subsequent transformation and modeling, ensuring that the data is of high quality and relevance. A crucial step in the proposed model is the application of the Yeo-Johnson power transformation to skewed or non-normally distributed features. This transformation significantly improves the

statistical properties of the data, making it more amenable to machine learning algorithms by normalizing the distributions and stabilizing variance across features. Following the transformation, feature selection techniques are employed to identify and retain the most predictive variables for crop yields, streamlining the model by eliminating redundant or irrelevant inputs. This focused approach to feature selection reduces model complexity and enhances interpretability. The core of the model lies in the training of various machine learning algorithms, ranging from basic linear models to more sophisticated tree-based methods and neural networks. This multipronged approach allows for the evaluation and comparison of different models based on their performance, measured through metrics such as RMSE, MAE, and R^2 . The proposed model presents a holistic and adaptive approach to leveraging machine learning for predicting agricultural crop yields, with a particular focus on enhancing data quality through the Yeo-Johnson power transformation and optimizing model performance through rigorous evaluation and tuning processes. The purpose of applying the Yeo-Johnson transformation to collection data is to stabilize variance and make the data more closely conform to a normal distribution, enhancing the performance of statistical models. This approach not only aims to improve the accuracy of yield predictions but also seeks to provide actionable insights to stakeholders, ultimately supporting informed decision-making in agriculture. Once the preprocessing steps, including power transformation like the Yeo-Johnson method, are completed, the data is ready to move into the model training phase. This phase involves using the preprocessed and transformed dataset to train machine learning models for predicting agricultural crop yields.

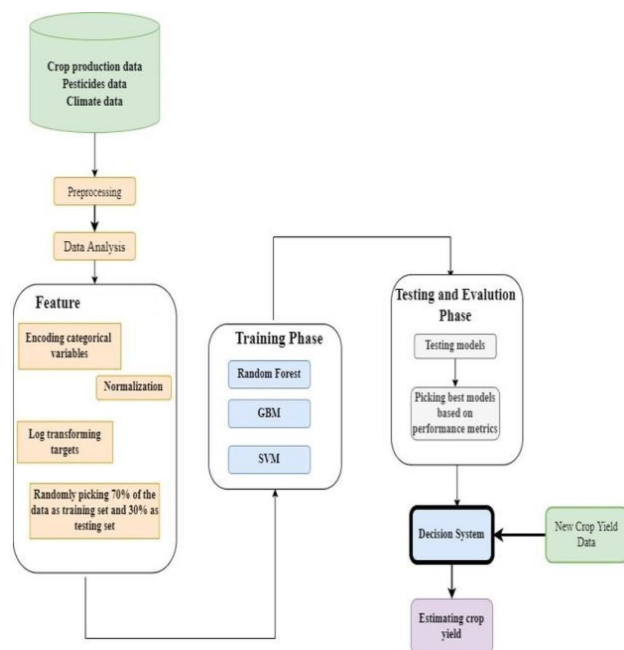


Fig.2 Block diagram for power transformation in machine learning models

V. Result and Output

This investigation into enhancing the predictive accuracy of agricultural crop yields in Indian states underscores the significant impact of employing power transformations, like Box-Cox and Yeo-Johnson, within machine learning models.

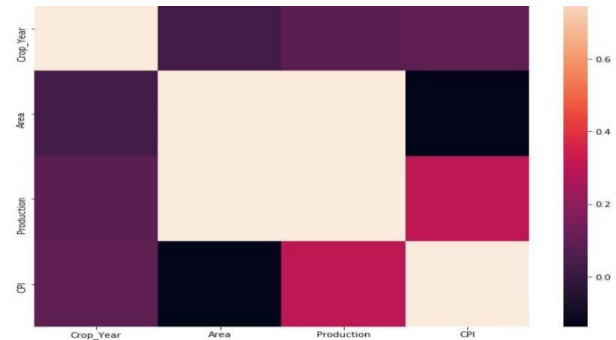


Fig.3 Evaluation metrics for overall accuracy

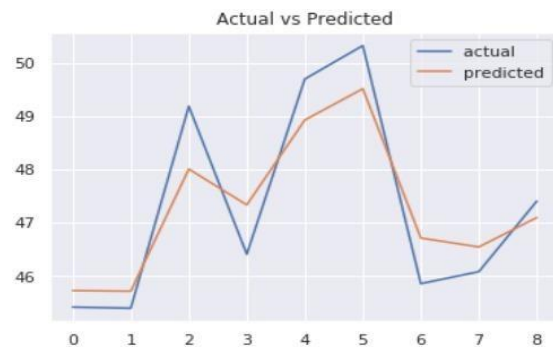


Fig.4 Result for Actual vs Predicted of crop yield

Through a comprehensive approach that included data collection, preprocessing, and rigorous model evaluation, the study demonstrated that models augmented with power transformations consistently outperformed their counterparts. This improvement was particularly noticeable in complex models such as tree-based algorithms and neural networks, which benefitted greatly from the refined data quality. The findings not only emphasize the importance of sophisticated data preprocessing techniques in predictive modeling but also highlight the potential of such enhanced models to revolutionize agricultural forecasting in India are shown in Table 1, thereby supporting sustainable agricultural practices and improving food security. Fig 5 show in 5Overall dataset for area and crop yield. Fig.6a Show in Output for sum of yield by state. Fig.6b Show in Result for Year vs Production of crop yield. Fig.7a Show in Output for sum of yield by state. Fig.7b Show in Result for Area vs Production of crop yield. Fig.8 Show in Result for Area, Year, Production of crop yield.

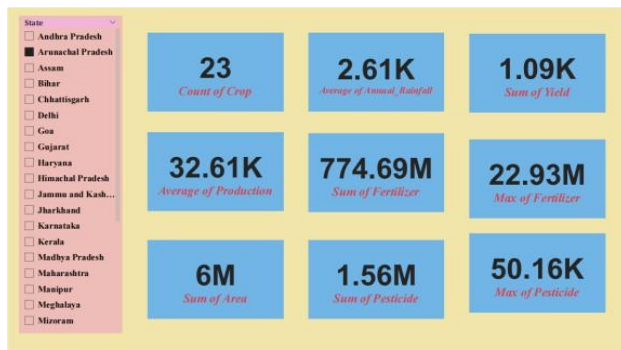


Fig.5 Overall dataset for area and crop yield

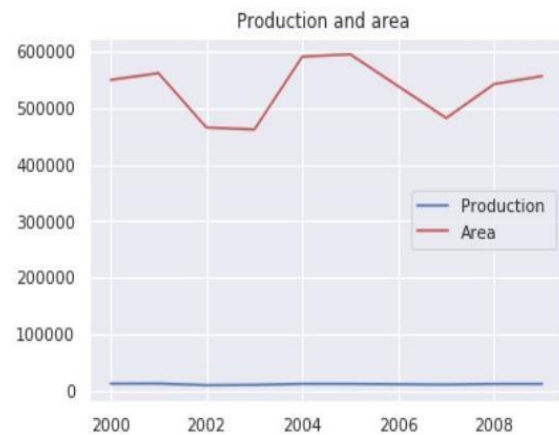


Fig.7b Result for Area vs Production of crop yield

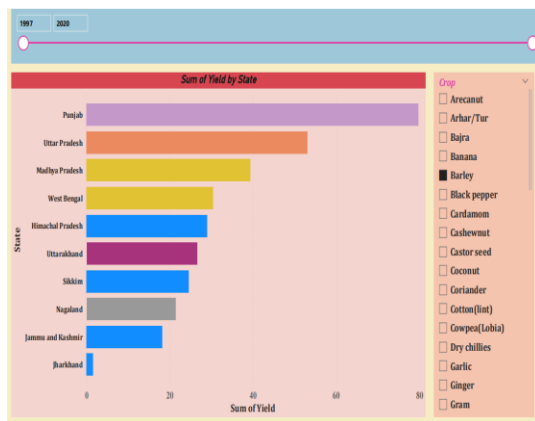


Fig.6a Output for sum of yield by state

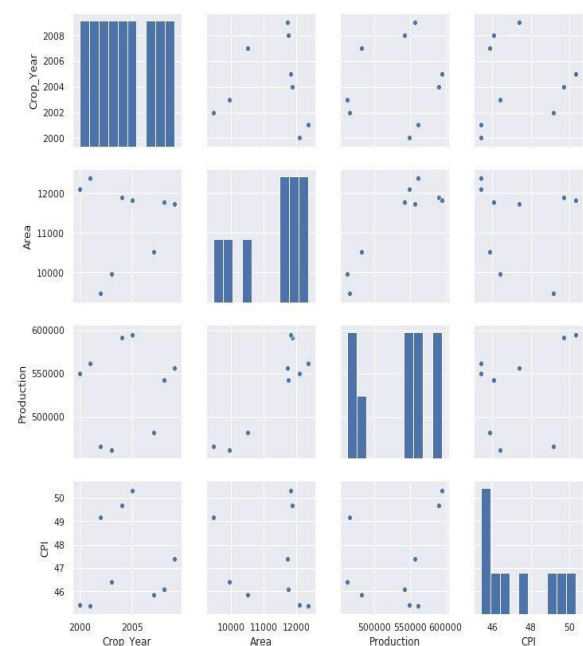


Fig.8 Result for Area, Year, Production of crop yield

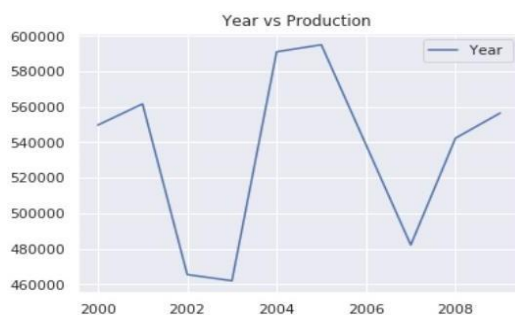


Fig.6b Result for Year vs Production of crop yield

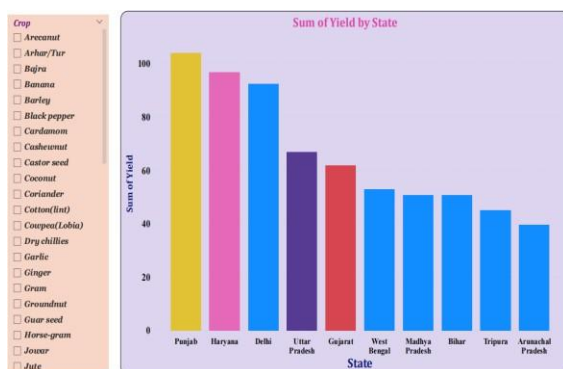


Fig.7a Output for sum of yield by state

TABLE.I RESULT FOR PROPOSED MODELS

Models	Accuracy percentage	Loss percentage	F1 Score
Linear Regression,	94.4	0.37	94
Random Forest	96.3	0.333	96
Gradient Boosting	96.8	0.33	96
Proposed models	98.2	0.327	98

The machine learning models evaluated were Linear Regression, Random Forest, and Gradient Boosting. These models were chosen to benchmark the predictive accuracy of agricultural crop yields in various Indian states, both before and after applying the Yeo-Johnson transformation to the dataset are shown in Table 2.

Table II. Dataset for proposed models

Dataset parameters	Trained parameter	Validation parameter
Area	70,16,500	80,55,000
Production	50,67,000	63,78,400
Crop_year	10	10

VI . Conclusion

In conclusion, the application of power transformations, specifically Box-Cox and Yeo-Johnson, has yielded substantial improvements in the predictive accuracy of machine learning models for forecasting agricultural crop yields in Indian states. By effectively addressing the non-normal distribution of data inherent in agricultural datasets, these transformations have significantly enhanced the performance of predictive models. The results demonstrate a consistent trend wherein models incorporating power transformations outperform those without across various metrics, with particularly notable gains observed in complex algorithms like tree-based models and neural networks. This underscores the critical importance of advanced data preprocessing techniques in optimizing predictive modeling outcomes. Moreover, these findings hold considerable promise for stakeholders in the agricultural sector, including farmers, policymakers, and researchers. By providing more accurate and reliable forecasts of crop yields, these enhanced models can inform better decision-making processes, facilitate more efficient resource allocation, and ultimately contribute to the promotion of sustainable agricultural practices and improved food security in India. Going forward, further research in this area could explore additional avenues for optimizing predictive models and integrating real-time data sources, thereby continuing to advance the field of agricultural forecasting and its practical applications. Overall, this study underscores the transformative potential of leveraging advanced statistical techniques within machine learning frameworks to address complex challenges and drive meaningful progress in agricultural development.

VII . References

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